Module 1: Introduction to Artificial Intelligence

Detailed Instructor Script / Speaker Notes

Slide 1 — Introduction to Artificial Intelligence

Welcome everyone. Today we are beginning Module 1: Introduction to Artificial Intelligence. This module is designed to help you understand what Artificial Intelligence, or AI, is, how it developed, how it appears in everyday life, and why it matters in STEM fields.

AI is no longer something that only computer scientists or researchers need to understand. It now affects engineering, healthcare, physics, biology, cybersecurity, transportation, manufacturing, environmental science, and many other areas. By the end of this module, you should have a strong beginner-level foundation that will help you understand later topics in this course.

As we move through the lecture, I want you to think about two questions. First, where do you already see AI in your daily life? Second, how might AI affect your future career?

Transition: Let’s begin by thinking about what comes to mind when we hear the phrase Artificial Intelligence.

Slide 2 — Think About It...

Before we define Artificial Intelligence, I want you to pause and think about what the phrase means to you.

When you hear “Artificial Intelligence,” what comes to mind? Some people immediately think about robots. Others think about ChatGPT, self-driving cars, smart assistants like Alexa or Siri, or even science fiction movies.

Take about one minute and discuss with a classmate: What is the first example of AI that comes to your mind?

Possible student answers may include ChatGPT, robots, Tesla vehicles, Alexa, Siri, Google Maps, Netflix, facial recognition, or movie characters like Terminator or JARVIS.

The important point is that many of our ideas about AI come from both real technologies and fictional stories. Some of those ideas are accurate, and some are not.

Transition: Now that we have identified what AI makes us think of, let's separate the myths from the reality.

Slide 3 — AI in Pop Culture: Myth vs Reality

This slide helps us separate what popular culture says about AI from what AI actually is.

Many movies and shows make AI seem conscious, emotional, dangerous, or human-like. They may show AI wanting to take over the world, thinking exactly like a human, or being represented only as a robot.

In reality, current AI does not have consciousness, feelings, goals, or desires. AI systems use mathematics, data, algorithms, and computing power. They learn patterns from data and use those patterns to produce predictions, recommendations, classifications, or generated content.

A major misconception is that AI is always accurate. That is false. AI can make mistakes. It can produce incorrect answers, reflect bias in training data, or misunderstand the context of a question.

Student question: Why do you think movies often exaggerate AI?

Expected answer: Because dramatic, human-like AI makes stories more exciting, but it does not represent how most real AI systems work today.

Transition: Now let's connect AI to why this matters specifically for STEM students.

Slide 4 — Why Should STEM Students Learn AI?

AI literacy is becoming as important as computer literacy. In the past, being able to use a computer was considered a special skill. Today, computer literacy is expected in almost every career. AI literacy is moving in the same direction.

AI is transforming engineering, healthcare, physics, biology, cybersecurity, environmental science, manufacturing, and many other fields. Engineers use AI to design smarter systems. Healthcare professionals use AI to support diagnosis and patient monitoring. Environmental scientists use AI to study climate patterns, wildlife, pollution, and natural disasters.

This does not mean every STEM student must become an AI engineer. It means that every STEM student should understand what AI can do, what its limitations are, and how to use it responsibly.

Discussion question: How might AI affect your future career?

Expected answers may include automation, better data analysis, faster research, improved medical tools, smarter machines, cybersecurity tools, or new career opportunities.

Transition: Let's look at what you should be able to do by the end of this module.

Slide 5 — Learning Objectives

These are the main learning objectives for Module 1.

By the end of this module, you should be able to define Artificial Intelligence, explain major milestones in AI history, distinguish between AI, Machine Learning, and Deep Learning, describe intelligent agents, explain Artificial General Intelligence, identify AI applications in STEM, and discuss ethical concerns surrounding AI.

These objectives are important because they give us the foundation for the rest of the course. Later modules will go deeper into how AI systems represent knowledge, reason, handle uncertainty, learn from data, use neural networks, process language, and generate content.

For now, the goal is not to become an AI expert in one day. The goal is to build a clear foundation so that the later technical ideas make sense.

Transition: Let's begin with the most important question: What is Artificial Intelligence?

Slide 6 — What Is Artificial Intelligence?

Artificial Intelligence is the ability of computer systems to perform tasks that normally require human intelligence.

These tasks may include learning, reasoning, problem solving, pattern recognition, language understanding, and decision making.

For example, when a phone recognizes your face, that involves pattern recognition. When Google Maps suggests a faster route, that involves prediction and decision support. When ChatGPT answers a question, that involves language processing and text generation.

It is important to understand that AI does not think exactly like a human. Instead, AI systems use data and algorithms to identify patterns and produce useful outputs.

Student question: Which example on this slide sounds most like something humans do naturally?

Expected answers: Learning, reasoning, problem solving, language understanding, or decision making.

Transition: Now let’s compare human intelligence and artificial intelligence more directly.

Slide 7 — Human Intelligence vs Artificial Intelligence

This slide compares human intelligence and artificial intelligence.

Humans learn from experience, use intuition, understand context, show creativity, use common sense, and understand emotions. Humans can often understand meaning, humor, tone, and social situations even when the information is incomplete.

AI works differently. AI learns from data, recognizes patterns, makes predictions, processes large amounts of information quickly, and can perform certain tasks with great speed and consistency.

The figure compares the human brain and an AI system. On the human side, we see strengths such as creativity, flexibility, emotional understanding, and context. On the AI side, we see strengths such as speed, data processing, pattern recognition, and consistency.

Key point: AI can be powerful, but it is not human intelligence. It does some tasks extremely well, but it lacks human common sense and lived experience.

Transition: Now let’s look at AI systems you probably use every day.

Slide 8 — Everyday AI

This slide shows examples of AI systems that many people interact with every day.

Google Maps uses AI to predict traffic and recommend routes. Netflix recommends shows based on viewing behavior. Spotify suggests music. Siri and Alexa process voice commands. Face ID recognizes your face. ChatGPT generates text responses.

Ask students: Which AI system on this slide do you use most often?

Expected answers may include Google Maps, Netflix, Spotify, Siri, Alexa, Face ID, or ChatGPT.

The key idea is that AI is often invisible. Students may not always notice they are using AI because it is built into apps, websites, phones, cars, and digital services.

Transition: Let’s expand this idea and look at how AI is already embedded in many parts of society.

Slide 9 — AI Is Already Everywhere

AI powers many systems that people use without thinking about them as AI.

Search engines use AI to rank results. Social media feeds use AI to decide what content appears first. Online shopping systems recommend products. Healthcare systems may use AI to analyze medical images or patient data. Banks use AI for fraud detection. Smart devices use AI to respond to voice commands or adjust settings.

A good real-world example is Amazon’s recommendation engine. When Amazon suggests products you might want, it is using past behavior, browsing history, purchases, and patterns from other users to make predictions.

Student question: Is a recommendation always correct?

Expected answer: No. AI makes predictions based on data, but predictions can be wrong or incomplete.

Transition: Since AI is everywhere, we also need to be clear about what AI is not.

Slide 10 — What AI Is NOT

This is an important slide because many students come into AI courses with misconceptions.

AI is not magic. It does not work through mysterious thinking. It is built from software, data, mathematics, algorithms, and computing power.

AI is not conscious. It does not have human feelings or awareness. AI is not human, even when it produces human-like language. AI is not perfect. It can make mistakes, reflect bias, or generate inaccurate information. AI is also not always a robot. Many AI systems are invisible software systems running in the background.

Key idea: AI is a tool. Like any tool, it can be useful or harmful depending on how it is designed and used.

Transition: Now that we understand what AI is and is not, let's look at its benefits.

Slide 11 — Benefits of AI

AI can bring many benefits when it is designed and used responsibly.

AI can improve efficiency by automating repetitive tasks. It can increase productivity by helping people complete work faster. In healthcare, AI can help detect diseases from medical images. In transportation, AI can improve safety by identifying risks. In scientific research, AI can accelerate discoveries by analyzing large datasets. AI can also support decision-making by identifying patterns that humans may miss.

The figure on this slide shows several benefits, including efficiency, accuracy, better decisions, innovation, safety, and quality of life.

Real-world example: AI can help doctors review X-rays or MRIs more quickly, but the doctor still uses medical judgment.

Transition: AI has benefits, but it also creates challenges. Let's look at those next.

Slide 12 — Challenges of AI

AI can also create serious challenges.

AI systems can produce biased results if the data used to train them contains bias. They can create privacy concerns when large amounts of personal data are collected. They can introduce security risks if systems are attacked or misused. AI can also contribute to misinformation, especially through deepfakes or automatically generated false content. AI may also disrupt jobs by changing the kinds of tasks people do.

The figure shows a balance between opportunities and risks. This is important. The goal is not to avoid AI completely. The goal is to build and use AI responsibly.

Student question: Which challenge concerns you the most?

Expected answers may include privacy, bias, job loss, misinformation, or safety.

Transition: To understand where AI is going, it helps to understand where AI came from.

Slide 13 — Where Did AI Come From?

Before we understand modern AI, we need to look briefly at its history.

AI did not appear suddenly with ChatGPT. The idea of machine intelligence has existed for many decades. Researchers began asking whether machines could reason, solve problems, play games, understand language, and act intelligently long before today's tools existed.

The timeline figure shows that AI developed through stages: early theoretical ideas, early programs, periods of progress, periods of disappointment, machine learning breakthroughs, and modern generative AI.

Key idea: AI is not brand new. What is new is the scale, power, accessibility, and visibility of modern AI tools.

Transition: We begin this history with Alan Turing.

Slide 14 — Alan Turing

Alan Turing is often called the Father of Modern Computing. In 1950, he published an influential paper titled Computing Machinery and Intelligence.

In that paper, Turing asked a question that became foundational for AI: Can machines think?

This was a powerful question because it shifted attention from only building machines that calculate to imagining machines that might imitate intelligent behavior.

The figure shows Alan Turing and highlights his role as a mathematician, logician, and pioneer of computing. His ideas helped shape computer science, artificial intelligence, and modern technology.

Student question: Why do you think asking "Can machines think?" was such a bold question in 1950?

Expected answer: Computers were very limited at the time, so imagining machine intelligence was extremely forward-thinking.

Transition: Turing proposed a way to test machine intelligence, which we now call the Turing Test.

Slide 15 — The Turing Test

The Turing Test is one of the earliest ways people tried to evaluate machine intelligence.

A machine passes the Turing Test if a human cannot reliably distinguish between the machine and another human through conversation.

This does not mean the machine truly understands like a human. It means the machine can imitate human-like conversation well enough to confuse a human judge.

Activity question: Would ChatGPT pass the Turing Test?

Possible answer: In some situations, ChatGPT may appear very human-like in conversation. However, passing a conversation test does not prove that ChatGPT is conscious or that it understands like a human.

Key point: The Turing Test is historically important, but it is not the only way to evaluate AI.

Transition: A few years after Turing's paper, AI became an official field of study.

Slide 16 — Birth of Artificial Intelligence

The term Artificial Intelligence became widely associated with a 1956 event called the Dartmouth Conference.

At this conference, researchers gathered to study whether machines could simulate aspects of intelligence. John McCarthy is credited with coining the term Artificial Intelligence.

This moment is often described as the official birth of AI as a field.

The historical timeline graphic helps students see that AI began as a research vision before it became a daily technology.

Key idea: AI started as a research question. Today, it has become a major part of industry, education, science, healthcare, and everyday life.

Transition: Let's look at some major milestones in AI history.

Slide 17 — AI Timeline

This timeline shows major moments in AI history.

In 1950, Alan Turing asked whether machines could think. In 1956, the Dartmouth Conference helped launch AI as a formal field. In 1966, ELIZA became one of the first famous chatbots. In 1997, IBM's Deep Blue defeated chess champion Garry Kasparov. In 2011, IBM Watson won Jeopardy. In 2012, AlexNet helped launch the modern deep learning revolution. In 2016, AlphaGo defeated Go champion Lee Sedol. In 2022 and beyond, tools like ChatGPT brought generative AI into public awareness.

Point to the timeline and emphasize that AI progress has not been a straight line. There were breakthroughs, setbacks, and renewed interest.

Transition: Let's look more closely at one early chatbot: ELIZA.

Slide 18 — ELIZA: The First Famous Chatbot

ELIZA was created in 1966 and simulated a therapist-style conversation.

It was not intelligent in the way modern AI systems are. It often used simple pattern matching and scripted responses. However, it was important because it showed that people could feel like they were interacting with an intelligent system even when the system was relatively simple.

Many people assume conversational AI began recently with ChatGPT. ELIZA shows that the idea of chatbots is almost 60 years old.

Student question: Why might people feel that ELIZA was intelligent, even if it used simple rules?

Expected answer: Because conversation feels human, and people may project understanding onto a system that responds in human-like ways.

Transition: AI also became famous through game-playing systems, especially chess.

Slide 19 — Deep Blue vs Kasparov

In 1997, IBM's Deep Blue defeated Garry Kasparov, the world chess champion.

This was a major milestone because chess had long been viewed as a symbol of human intelligence, strategy, and planning. Deep Blue showed that a machine could outperform humans in a specialized intellectual task.

The figure shows the symbolic matchup between a human chess champion and a machine. Emphasize that Deep Blue was powerful at chess, but it was not generally intelligent. It could not drive a car, write an essay, or diagnose a disease.

Key idea: AI can be extremely strong in a narrow task without being generally intelligent.

Transition: Another major AI milestone happened on the quiz show Jeopardy.

Slide 20 — IBM Watson

In 2011, IBM Watson won Jeopardy against top human champions.

This was impressive because Jeopardy requires more than memorizing facts. It involves natural language, wordplay, clues, categories, timing, and retrieving information quickly.

Watson demonstrated Natural Language Processing, question answering, and knowledge retrieval.

The figure shows Watson competing in a game-show environment. Explain that Watson was designed for this specific task. It did not understand the world like a human, but it could process questions and search for likely answers very quickly.

Student question: Why is Jeopardy harder for AI than chess?

Expected answer: Because language is ambiguous. Words can have multiple meanings, and clues may involve humor, context, or wordplay.

Transition: Next, we move from chess and Jeopardy to the game of Go.

Slide 21 — AlphaGo

In 2016, AlphaGo defeated world champion Lee Sedol in the game of Go.

This was a major AI milestone because Go is much more complex than chess. The number of possible moves is enormous, and expert play often involves intuition and strategy.

The AlphaGo figure shows how this system combined learning from games with advanced search techniques. For Module 1, we do not need to understand the technical details. The key point is that AI began solving problems that were once considered too complex for machines.

AlphaGo also showed that AI could sometimes discover strategies that surprised human experts.

Transition: These historical milestones helped lead to today's modern AI revolution.

Slide 22 — Modern AI Revolution

Modern AI has been powered by several major developments: Big Data, GPUs, cloud computing, deep learning, and access to large-scale digital information.

Big Data means AI systems can train on massive datasets. GPUs provide the computing power needed to process that data. Cloud computing makes powerful AI tools available online. Deep learning allows models to learn complex patterns from images, sound, text, and other data.

Generative AI tools include ChatGPT, Gemini, Claude, DALL-E, and Midjourney. These tools can generate text, images, code, and other content.

Key idea: Modern AI became powerful because data, computing power, and algorithms all improved at the same time.

Transition: Now we need to clarify an important distinction: not all AI is the same.

Slide 23 — Not All AI Is the Same

AI, Machine Learning, and Deep Learning are related concepts, but they are not identical.

Many people use these terms interchangeably, but in AI education it is important to understand the difference.

Artificial Intelligence is the broadest category. Machine Learning is a subset of AI. Deep Learning is a subset of Machine Learning.

This distinction matters because later modules will use these terms more precisely. If students confuse AI, ML, and DL now, later topics will become harder.

Transition: The next slide gives us one of the most important visuals in the module.

Slide 24 — AI vs ML vs DL

This slide explains the relationship between Artificial Intelligence, Machine Learning, and Deep Learning.

Artificial Intelligence is the broad field. It includes any system designed to perform tasks that normally require human intelligence.

Machine Learning is inside AI. It focuses on systems that learn from data instead of being explicitly programmed for every rule.

Deep Learning is inside Machine Learning. It uses neural networks with many layers to learn complex patterns.

Point to the nested-circle or hierarchy diagram. Explain that all Deep Learning is Machine Learning, and all Machine Learning is AI, but not all AI is Machine Learning, and not all Machine Learning is Deep Learning.

Key takeaway: AI is the big field. ML is inside AI. DL is inside ML.

Transition: Let’s look at each term separately.

Slide 25 — Artificial Intelligence

Artificial Intelligence is the broad field focused on creating intelligent systems.

AI can include expert systems, search algorithms, robotics, planning systems, natural language systems, and Machine Learning.

This is important because AI existed before modern Machine Learning became popular. Early AI systems often used rules written by humans. For example, an expert system might contain rules such as: If a patient has certain symptoms, then suggest possible conditions.

Today, many AI systems use Machine Learning, but AI is still broader than Machine Learning.

Real-world example: A robot that follows programmed rules to move through a warehouse may be considered AI even if it is not learning from data.

Transition: Now let’s focus on Machine Learning.

Slide 26 — Machine Learning

Machine Learning is a subset of AI in which computers learn patterns from data.

Instead of programming every rule by hand, we provide data, and the system learns patterns from that data.

For example, a spam filter learns from many examples of spam and non-spam emails. A fraud detection system learns from past transactions. A recommendation system learns from user behavior.

Machine Learning is powerful because it can discover patterns that may be too complex for humans to write as rules.

Key idea: Machine Learning allows computers to improve performance by learning from examples.

Transition: Deep Learning is a more specialized type of Machine Learning.

Slide 27 — Deep Learning

Deep Learning is a type of Machine Learning that uses artificial neural networks to identify complex patterns.

Deep Learning is especially useful for large, complex data such as images, speech, and language.

Examples include face recognition, speech recognition, generative AI, self-driving vehicles, and medical image analysis.

For this module, we are keeping Deep Learning high level. Students do not need to understand weights, backpropagation, gradient descent, or activation functions yet. Those are more advanced topics for later modules.

Key idea: Deep Learning is powerful for recognizing complex patterns in large amounts of data.

Transition: Let’s use a real-world example to see AI, ML, and DL together.

Slide 28 — Real-World Example: Self-Driving Car

A self-driving car is a useful example because it combines many AI technologies.

The entire driving system is AI. It includes sensing, planning, decision-making, navigation, and control.

Machine Learning may be used to learn driving patterns, predict traffic behavior, and improve performance over time.

Deep Learning may be used to recognize objects such as pedestrians, traffic signs, lane markings, cyclists, and other vehicles.

Point to the layered car diagram. Explain that different layers work together: sensors collect information, perception systems interpret the environment, planning systems choose actions, and control systems steer, brake, or accelerate.

Key idea: Real AI systems often combine multiple technologies.

Transition: Let's check understanding before we continue.

Slide 29 — Knowledge Check

Ask students: Which is the broadest concept?

A. Deep Learning
B. Machine Learning
C. Artificial Intelligence

Give students a few seconds to answer.

Correct answer: C. Artificial Intelligence.

Explanation: Artificial Intelligence is the broadest field. Machine Learning is a subset of Artificial Intelligence. Deep Learning is a subset of Machine Learning.

Follow-up question: If a system uses Deep Learning, is it also Machine Learning?

Expected answer: Yes.

Follow-up question: If a system uses Machine Learning, is it always Deep Learning?

Expected answer: No.

Transition: Now that we understand AI, ML, and DL, let's learn important vocabulary that will help us understand future modules.

Slide 30 — Before We Continue...

Before we move into intelligent agents and applications, we need to learn some important AI vocabulary.

The terms algorithm, data, training, model, neural network, Natural Language Processing, and Computer Vision appear throughout AI discussions. Students do not need to master every technical detail today, but they should understand the basic meaning of each term.

This vocabulary will also help students complete the AI Definitions Scavenger Hunt and the AI vs ML vs DL Visual Essay.

Transition: Let's begin with one of the most important words in computer science and AI: algorithm.

Slide 31 — Algorithm

An algorithm is a finite, step-by-step procedure used to solve a problem or achieve a goal.

A simple analogy is a recipe. A recipe has ingredients, steps, a process, and a final dish. An algorithm works similarly. It takes input, follows steps, processes information, and produces output.

For example, a GPS navigation algorithm may take your current location and destination as input. It then calculates possible routes and outputs the best route based on distance, traffic, and time.

Point to the flowchart. Explain that flowcharts show how algorithms make decisions step by step.

Key idea: Algorithms break problems into clear, ordered steps so computers can solve them efficiently.

Transition: Algorithms often need data. Let's define data next.

Slide 32 — Data

Data means raw facts, numbers, text, images, audio, video, or sensor readings that can be recorded, stored, and analyzed.

AI systems need data because data is how they learn patterns.

The figure gives several examples of data: numerical data, text data, image data, audio data, video data, and sensor or time-series data.

A real-world example is Netflix viewing history. Netflix collects data about what shows you watch, how long you watch, what you pause, what you skip, and what you rate. It then uses that data to recommend shows and movies.

Student question: What kind of data does Spotify collect?

Expected answers: Songs listened to, skipped songs, playlists, likes, listening time, genre preferences.

Transition: Once we have data, AI systems can be trained.

Slide 33 — Training

Training is the process of teaching an AI model using data so it can learn patterns and make predictions.

Use the cat vs dog example. Suppose we want to train an AI system to recognize cats and dogs. We provide many labeled images. Some images are labeled “cat,” and others are labeled “dog.” The system looks for patterns such as ear shape, face shape, fur texture, eye position, and body shape.

During training, the system makes predictions, checks whether it was correct, and adjusts itself to reduce errors. Over time, it gets better at classifying new images.

Point to the training process infographic. Emphasize that training requires data, labels, learning patterns, testing, and improvement.

Key idea: Training is how AI learns from examples.

Transition: After training, the result is called a model.

Slide 34 — Model

A model is an AI system that has learned patterns from data and can use those patterns to make predictions, classifications, recommendations, or generate new content.

Think of a model as the trained result of the learning process.

A simple analogy is a student practicing many math problems before an exam. After enough practice, the student recognizes patterns and solves new problems. An AI model works similarly: it studies examples, learns patterns, and applies those patterns to new situations.

A real-world example is ChatGPT. ChatGPT is a language model. It was trained on large amounts of text and learned patterns in language. It can answer questions, summarize information, write essays, generate code, and hold conversations.

Important: ChatGPT does not understand like a human. It predicts likely words based on learned patterns.

Transition: Many modern models use neural networks.

Slide 35 — Neural Networks

A neural network is a computing system inspired by the human brain, but it is not an actual brain.

A neural network is made of layers of connected units called nodes or artificial neurons. These nodes process information and pass signals through the network. The network learns patterns from data during training.

Point to the neural network diagram. The input layer receives data. The hidden layers process patterns. The output layer produces a result, such as a prediction or classification.

Keep this explanation high level. We are not teaching the mathematics of neural networks in Module 1.

Real-world examples include image recognition, speech recognition, language generation, and recommendation systems.

Key idea: Neural networks help AI systems recognize complex patterns.

Transition: One area where AI uses neural networks and other methods is language.

Slide 36 — Natural Language Processing

Natural Language Processing, or NLP, is the area of AI that helps computers work with human language.

Human language is complicated. Words can have multiple meanings. Sentences depend on context. Tone, grammar, slang, and culture all affect meaning.

Examples of NLP include ChatGPT, Siri, and Google Translate. ChatGPT generates text responses. Siri processes spoken commands. Google Translate converts text from one language to another.

Student question: Why is human language difficult for computers?

Expected answer: Language can be ambiguous, context-dependent, emotional, informal, and full of multiple meanings.

Key idea: NLP allows computers to process, interpret, and generate human language.

Transition: Another major area of AI helps machines interpret images and video.

Slide 37 — Computer Vision

Computer Vision is the area of AI that helps machines interpret visual information such as images and videos.

Examples include Face ID, medical imaging, and self-driving vehicles.

Point to the image classification figure. Explain that a computer vision system receives an image, extracts visual features, uses a trained model, and produces a prediction. For example, it may classify an image as a dog, cat, car, or bird.

In healthcare, computer vision may help identify abnormalities in X-rays or scans. In self-driving vehicles, computer vision helps detect pedestrians, traffic signs, lanes, and other vehicles.

Important: Computers do not “see” like humans. They analyze pixels, patterns, shapes, and learned features.

Transition: Now we will connect these ideas to intelligent agents.

Slide 38 — What Is an Intelligent Agent?

An intelligent agent is a system that observes what is happening around it, decides what to do, and then acts.

An intelligent agent can perceive its environment, process information, make decisions, and take actions to achieve a goal.

Examples include Roomba, Alexa, and Google Maps.

A Roomba senses dirt, walls, and furniture, then decides where to clean. Alexa hears a voice command, processes the request, and gives a response. Google Maps detects traffic conditions, calculates routes, and recommends the best path.

Key idea: An intelligent agent does not just store information. It uses information to make decisions and act.

Student question: What makes Google Maps more than just a digital map?

Expected answer: It processes live information, makes route decisions, and updates recommendations.

Transition: Intelligent agents often follow a cycle.

Slide 39 — Agent Cycle

The agent cycle shows how an intelligent agent continuously interacts with its environment.

The cycle begins with the environment. The agent perceives information using sensors or input. Then it thinks by processing that information. It decides what action to take. It acts in the environment. It learns from the result. Then the cycle repeats.

Point to the circular diagram. Explain that this is a loop, not a one-time process.

Ask students: Which step is most important?

Expected answer: All of them are important. If an agent cannot perceive, it does not know what is happening. If it cannot decide, it cannot act intelligently. If it cannot act, it cannot affect the environment. If it cannot learn, it cannot improve.

Transition: Next, let’s break the agent into its major components.

Slide 40 — Components of an Agent

An intelligent agent usually has three major components: sensors, a processor, and actuators.

Sensors collect information from the environment. A robot may use cameras, microphones, distance sensors, temperature sensors, or GPS.

The processor interprets the information, makes decisions, and plans action. This is sometimes described as the “brain” of the agent, though it is not a human brain.

Actuators carry out actions. These may include wheels, robotic arms, speakers, screens, motors, or software outputs.

Point to the robot architecture diagram. Show how information comes from the environment, is processed by the agent, and then leads to action.

Key idea: Agents perceive, decide, and act.

Transition: Let’s apply this idea in a class activity.

Slide 41 — Class Activity: Is It an Intelligent Agent?

For this activity, students will decide whether different systems are intelligent agents.

Ask students to use four questions:
Does the system perceive its environment?
Does it process information?
Does it make decisions?
Does it take actions?

Explain that not every technology is an intelligent agent. A calculator performs operations, but it does not perceive its environment or adapt. A Roomba, however, senses obstacles, processes information, and changes its behavior.

Have students work with a partner or small group for two minutes before reviewing the table.

Transition: Now let’s look at the table and discuss the answers.

Slide 42 — Class Activity Table

Review the table with students.

Roomba is an intelligent agent because it senses, decides, and acts. Alexa is an intelligent agent because it receives voice input, processes language, and responds. Google Maps is an intelligent agent because it uses location and traffic data to recommend routes. A calculator is not an intelligent agent because it follows fixed instructions and does not perceive or adapt.

A traffic light timer is tricky. It takes action, but if it is only a fixed timer, it does not truly perceive or adapt. A smart traffic light with sensors would be closer to an intelligent agent.

A thermostat is usually an intelligent agent because it senses temperature, decides whether heating or cooling is needed, and acts.

Student question: Is ChatGPT an intelligent agent?

Expected answer: Yes, in a limited sense, because it receives input, processes information, generates responses, and interacts with users.

Transition: Let’s discuss a few of these examples more deeply.

Slide 43 — Follow-Up Discussion

This slide gives students a chance to explain their reasoning.

Question 1: Why is a calculator not considered an intelligent agent?

Expected answer: It follows fixed instructions and does not perceive its environment or adapt.

Question 2: Why is a Roomba considered an intelligent agent?

Expected answer: It senses obstacles and dirt, makes decisions, and changes its behavior based on the environment.

Question 3: Is ChatGPT an intelligent agent?

Expected answer: Yes, because it receives input, processes information, generates responses, and interacts with users. However, it does not have physical sensors or actuators like a robot.

Challenge question: Is a thermostat an intelligent agent?

Expected answer: Yes, because it senses temperature, decides whether heating or cooling is needed, and acts.

Transition: Now that we understand agents, let’s discuss a bigger idea: AGI.

Slide 44 — What Is AGI?

Artificial General Intelligence, or AGI, is a hypothetical type of AI that could understand, learn, reason, and solve problems across many different domains at a level similar to or beyond a human being.

Unlike today's AI systems, AGI would not be limited to one task. It could learn and perform almost any intellectual task that a human can do.

This is very important: ChatGPT is not AGI. Siri is not AGI. Alexa is not AGI. Google Maps is not AGI. These are examples of Narrow AI.

Use the figure to compare Narrow AI, Human Intelligence, and AGI. Narrow AI exists today and performs specific tasks. Human intelligence exists today and works across many domains. AGI is a future possibility and does not currently exist.

Transition: Let's look more closely at current AI systems.

Slide 45 — AGI: Current Reality

Today's AI systems are examples of Narrow AI, also called Artificial Narrow Intelligence or ANI.

Narrow AI systems are very good at specific tasks. ChatGPT is good at language tasks. Siri and Alexa are good at voice interaction. Netflix recommendations are good at predicting content preferences. Google Maps is good at navigation. Face ID is good at recognizing a face.

However, these systems cannot do everything a human can do. Google Maps cannot diagnose a disease. Face ID cannot write a research paper. Netflix cannot drive a car.

Key idea: Current AI systems can be powerful without being generally intelligent.

Transition: So what would AGI be able to do if it existed?

Slide 46 — What Would AGI Be Able To Do?

An AGI system could potentially learn new subjects without being fully retrained, solve unfamiliar problems, transfer knowledge from one field to another, understand context deeply, adapt to new situations, and perform many different intellectual tasks.

For example, a human can learn math, drive a car, cook a meal, hold a conversation, teach a class, and solve a new problem. AGI would aim for this kind of broad adaptability.

Important fact: AGI does not currently exist. Researchers and companies are studying it, but today's AI systems are still examples of Narrow AI.

Student question: Why is transferring knowledge from one field to another difficult for AI?

Expected answer: Most AI systems are trained for specific tasks and do not have broad human-like understanding.

Transition: Let's imagine what AGI might look like through an example.

Slide 47 — AGI Real-World Example

Imagine an AI system that could diagnose diseases, design bridges, write software, teach a college course, drive a car, and conduct scientific research without being specially programmed for each individual task.

That would be much closer to AGI.

Today's systems can do some of these tasks in limited ways, but no current AI system can do all of them with human-like general understanding.

Key takeaway: Today's AI equals specialized expert. Future AGI would be a general problem solver.

Emphasize that this is a future research goal, not current reality.

Transition: This distinction helps us understand why movies often exaggerate AI.

Slide 48 — AGI Visual Comparison

Use this slide to reinforce the visual comparison between Narrow AI, Human Intelligence, and AGI.

Point to Narrow AI first. Explain that Narrow AI exists today and is designed for specific tasks. Examples include ChatGPT, Google Maps, Face ID, and Netflix.

Point to Human Intelligence. Explain that humans can learn continuously, transfer knowledge, understand context, and adapt to many situations.

Point to AGI. Explain that AGI would try to combine broad learning, adaptability, and problem-solving across many domains. However, AGI does not exist today.

Key idea: The difference between today's AI and AGI is one of the most important ideas in introductory AI literacy.

Transition: Let's connect AGI to how AI is shown in movies.

Slide 49 — AI in Movies vs Reality

Movies often show AI as human-like, emotional, dangerous, or fully independent.

Examples include Terminator, JARVIS, Ex Machina, and HAL 9000. These fictional systems often represent AGI or even superintelligence.

Reality is very different. Today's AI includes ChatGPT, Siri, Alexa, and Google Maps. These systems are useful, but they are specialized tools.

Ask students: Why do movies portray AI differently from reality?

Expected answer: Because dramatic AI makes stories more exciting, but real AI is usually more specialized and less human-like.

Instructor note: Keep this discussion focused. Do not let the conversation drift too far into AI doom scenarios.

Transition: Now let's shift from general concepts to AI applications in STEM.

Slide 50 — AI in Engineering

AI helps engineers design, build, monitor, and improve complex systems.

Real-world engineering examples include predictive maintenance, smart manufacturing, robotics, structural monitoring, and design optimization.

Predictive maintenance is one of the best examples. Instead of waiting for a machine to break, sensors collect data about vibration, temperature, pressure, or performance. AI analyzes the data and predicts when failure may happen. This helps companies repair machines before they fail.

Smart manufacturing uses AI to optimize production, reduce waste, and improve quality. Robotics uses AI to help machines sense, decide, and act.

Key idea: AI helps engineers move from reacting to problems after they happen to predicting and preventing problems before they happen.

Transition: AI is also transforming healthcare.

Slide 51 — AI in Engineering Visual Examples

Use this visual slide to reinforce three engineering examples: smart factories, wind turbine monitoring, and industrial robots.

In the smart factory example, AI helps optimize production, detect defects, reduce energy use, and improve quality.

In the wind turbine monitoring example, sensors collect data about vibration, temperature, and performance. AI detects unusual patterns and predicts maintenance needs before a turbine fails.

In the industrial robots example, AI vision systems help robots identify objects, position tools, and perform tasks more precisely and safely.

Key takeaway: AI enables engineers to build smarter, safer, more efficient, and more sustainable systems.

Transition: Next, let's look at healthcare applications.

Slide 52 — AI in Healthcare

AI supports healthcare professionals by helping analyze medical data faster and more accurately.

AI can help with cancer detection by identifying possible tumors in medical images. It can assist with X-rays, MRIs, CT scans, and pathology images. It can help researchers search for new medicines through drug discovery. It can support personalized medicine by helping match treatments to patient history or genetic data. It can also help monitor patients by detecting warning signs from oxygen levels, heart rate, or other health data.

Important note: AI does not replace doctors. It supports doctors by helping them notice patterns and make better-informed decisions.

Key idea: AI can improve healthcare, but human judgment, ethics, and patient safety remain essential.

Transition: AI is also used to study the environment.

Slide 53 — AI in Environmental Science

AI helps scientists study the environment by analyzing large amounts of data from satellites, sensors, weather stations, and field observations.

Climate modeling uses AI to help predict climate patterns and extreme weather events. Wildlife tracking uses AI to analyze camera images, animal sounds, and GPS data. Energy optimization helps buildings and power grids use energy more efficiently.

Wildfire prediction is a strong example. AI can analyze weather conditions, vegetation, satellite imagery, and historical fire data to identify high-risk areas and support early warning systems.

Disaster prediction helps with hurricanes, floods, wildfires, and droughts. Pollution monitoring helps detect patterns in air and water quality.

Key idea: AI helps environmental scientists protect people, wildlife, and natural resources.

Transition: These applications are creating new career opportunities.

Slide 54 — AI Careers

AI is creating career opportunities across STEM fields, not only in computer science.

AI literacy is becoming important for engineers, scientists, healthcare professionals, cybersecurity workers, business analysts, environmental scientists, educators, and many others.

Students do not all need to become AI engineers. However, almost every STEM career will require some understanding of AI tools, data, automation, and ethical use.

Discussion question: Which AI-related career connects most closely to your major or career goal?

Possible student answers may include AI engineer, data scientist, robotics engineer, cybersecurity analyst, health informatics specialist, or AI ethics specialist.

Transition: The next slide lists several AI-related careers and what they do.

Slide 55 — Examples of AI-Related Careers

Review the career table with students.

An AI Engineer builds AI systems and applications. A Machine Learning Engineer creates models that learn from data. A Data Scientist analyzes data to find patterns and insights. A Robotics Engineer designs robots that sense, decide, and act. An AI Researcher studies new AI methods and technologies. A Cybersecurity Analyst uses AI to detect threats and protect

systems. A Health Informatics Specialist uses data and AI to improve healthcare. An AI Ethics Specialist studies fairness, privacy, safety, and accountability.

Ask students: Which of these careers did you already know about? Which one is new to you?

Key idea: AI creates both technical and non-technical career pathways.

Transition: Because AI affects people's lives, we must also talk about ethics.

Slide 56 — Section 8: Ethics

This section introduces AI ethics.

Ethics asks how we should design, use, and evaluate AI systems in ways that protect people and society.

AI is not only a technical issue. It is also a human issue. AI systems can influence hiring, healthcare, education, transportation, policing, lending, and access to information.

Because AI can affect real people, we must ask whether systems are fair, transparent, private, accountable, and safe.

Transition: Let's look at why ethics matters.

Slide 57 — Why Ethics Matters

AI systems affect people's lives. Ethical AI helps ensure technology is used responsibly, fairly, and safely.

This slide introduces five major principles: fairness, transparency, privacy, accountability, and safety.

Fairness means AI should avoid discrimination and biased outcomes. Transparency means people should understand how and why decisions are made. Privacy means personal data should be protected. Accountability means people and organizations must take responsibility for AI systems. Safety means AI should be reliable and should avoid causing harm.

Ask students: Which ethical principle do you think is most important, and why?

Possible answers: privacy, because personal information matters; fairness, because biased systems can harm people; safety, because unsafe AI can affect health or transportation.

Transition: Now let’s summarize what we learned.

Slide 58 — Module 1 Summary: What We Learned Today

This slide reviews the major learning objectives.

Today, students learned to define Artificial Intelligence, explain major milestones in AI history, distinguish between AI, Machine Learning, and Deep Learning, identify common AI vocabulary, describe intelligent agents, explain the difference between Narrow AI and AGI, recognize AI applications across STEM, and discuss ethical considerations.

This summary matters because Module 1 is the foundation for the rest of the course.

Ask students: Which topic from today felt most familiar? Which topic felt most new?

Expected answers may include ChatGPT or everyday AI as familiar, and intelligent agents or AGI as new.

Transition: Let’s review the biggest ideas to remember.

Slide 59 — Big Ideas to Remember

There are four big ideas to remember from this module.

First, AI is already part of daily life. It appears in Google Maps, Netflix, Face ID, ChatGPT, search engines, and many other tools.

Second, AI learns from data. Data, training, and models are the foundation of modern AI systems.

Third, AI is powerful but not human. Current AI systems are specialized tools, not conscious minds.

Fourth, ethics matters. As AI becomes more powerful, society must ensure it is fair, transparent, safe, accountable, and respectful of privacy.

Key idea: AI is useful, but it must be understood and used responsibly.

Transition: Now let’s complete an exit ticket.

Slide 60 — Exit Ticket

Before leaving class, students should answer four questions.

Question 1: What is one example of AI that you use regularly?

Expected answers: ChatGPT, Google Maps, Netflix, Spotify, Face ID, Siri, Alexa.

Question 2: In your own words, what is the difference between AI, Machine Learning, and Deep Learning?

Expected answer: AI is the broad field, Machine Learning is AI that learns from data, and Deep Learning is a type of Machine Learning that uses neural networks.

Question 3: What is one ethical concern related to AI?

Expected answers: bias, privacy, misinformation, safety, accountability, job disruption.

Question 4: Which AI application or career interested you most today?

Transition: These questions connect directly to your assignments.

Slide 61 — Connecting Today's Learning to Your Assignments

This slide connects the lecture to the module assignments.

The AI Definitions Scavenger Hunt helps students apply the vocabulary from today's lecture. The AI vs ML vs DL Visual Essay helps students demonstrate the relationship between Artificial Intelligence, Machine Learning, and Deep Learning. The Everyday AI Discussion asks students to analyze how AI affects daily life. The Ethics in AI Discussion asks students to evaluate opportunities and risks associated with responsible AI development.

Explain that the assignments are not separate from the lecture. They are designed to help students practice and apply what they learned today.

Transition: Finally, let's preview the next module.

Slide 62 — Looking Ahead

Today we learned what AI is. In Module 2, we will begin exploring how AI systems think, reason, and make decisions.

The next module is Knowledge Representation and Reasoning.

Students will learn how machines represent knowledge, how AI systems make decisions, how rules and logic can support reasoning, and how these ideas form the foundation of intelligent behavior.

This is a natural next step. Module 1 answers: What is AI? Module 2 begins answering: How do machines think?

Transition: Let's close with the key takeaway from this module.

Slide 63 — Key Takeaway

AI is transforming science, engineering, healthcare, business, education, and society.

Understanding AI is no longer optional for STEM professionals. It is becoming an essential skill for the future.

Students do not need to become AI experts immediately. However, they do need to understand AI vocabulary, examples, limitations, ethical concerns, and career connections.

Final message to students: As you continue through this course, remember that AI is not magic and it is not human. It is a powerful set of tools created by people, trained on data, and used to solve real-world problems. Your responsibility as future STEM professionals is to understand these tools and use them thoughtfully.

End of Module 1.